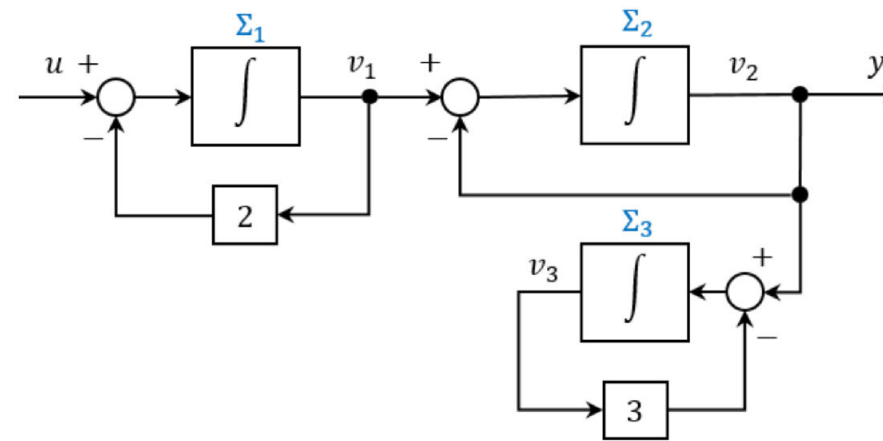


PART 1 - Problem 2

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC MATH TOOLBOX] [CONTROL TOOLBOX]

Consider the LTI system described by the block scheme in Figure



where Σ_1 , Σ_2 and Σ_3 are integrators with initial conditions

$$v_1(0) = +2.25, v_2(0) = 0, v_3(0) = -4.81$$

The input function is given by

$$u(t) = 5 e^{-4t} \cdot \mathbf{1}(t)$$

where $\mathbf{1}(t)$ denotes the unit step function.

Assignment:

- Determine a state-space representation of the LTI system.
- **[MANDATORY]** Sort the state variables in the state representation according to the labelling of the "integrator" systems, i.e. **firstly**, the state variables corresponding to Σ_1 , **then** the state variables associated with Σ_2 and **last**, the state variables corresponding to Σ_3 .
- Assign the matrices A , B , C and D of the obtained state representation to the **symbolic variables** **symA**, **symB**, **symC** and **symD**.
- Properly exploiting the Laplace transform, **determine analitically** the **output response** $y(t)$, given the assigned initial conditions and the input function $u(t)$. Store the resulting response in the **symbolic variable** **symYt**.

- Assign the matrices A , B , C and D of the obtained state representation to the **numeric variables** `matA`, `matB`, `matC` and `matD`.
- Using the matrices `matA`, `matB`, `matC` and `matD` define an LTI system description in state-space form, called `stateLTIsys`, exploiting the command `ss( )`. **Convert** this representation in a transfer function representation, exploiting the command `tf( )`. Assign the obtained transfer function to the variable `tfLTIsys`.
- Evaluate the eigenvalues of `matA` and store the result in the numeric array `eigA`.
- Evaluate zeros, poles and static gain of `tfLTIsys` and save the results in the **numeric variables** `zerosTF`, `polesTF` and `gainTF`.

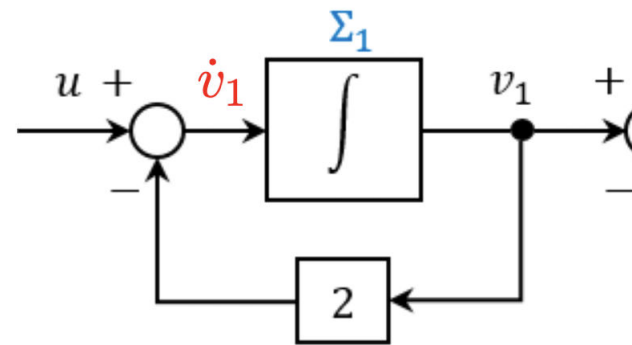
Solution

Analysis of the Scheme - The State-Space Equations of the Dynamic System

Let's consider the initial bullet points:

- **P1**: Determine a state-space representation of the LTI system.
- **P2: [MANDATORY] Sort** the state variables in the state representation according to the labelling of the "integrator" systems, i.e. **firstly**, the state variables corresponding to Σ_1 , **then** the state variables associated with Σ_2 and **last**, the state variables corresponding to Σ_3 .

Indeed the scheme depicted in the [Figure](#) is not an effective block diagram: it's a simple graphical representation of an Ordinary Differential Equation (ODE) of third order. Let's examine in detail the **first part of the scheme**, corresponding to the integrator with label Σ_1 :



Let's associate the variable $\dot{v}_1$ with the input branch of the "integrator" Σ_1 , as in the figure above (in red in the picture). Simply by inspection of the scheme, we can write

$$\dot{v}_1(t) = -2 v_1(t) + u(t)$$

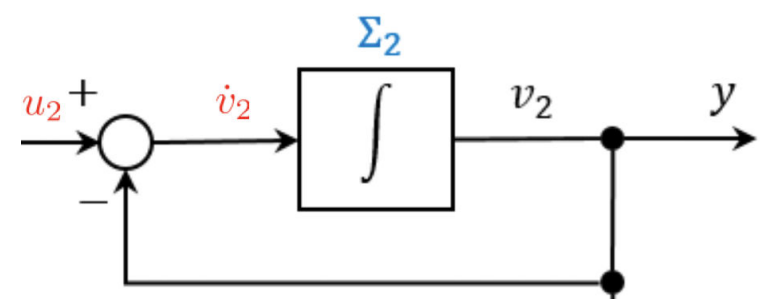
Moreover, considering the output branch of the integrator Σ_1 , the following relation holds

$$y_1(t) = v_1(t)$$

Thus, this portion of the scheme corresponds to the equations

$$\begin{cases} \dot{v}_1(t) &= -2 v_1(t) + u(t) \\ y_1(t) &= v_1(t) \end{cases}$$

Let's focus on **the central part of the scheme**

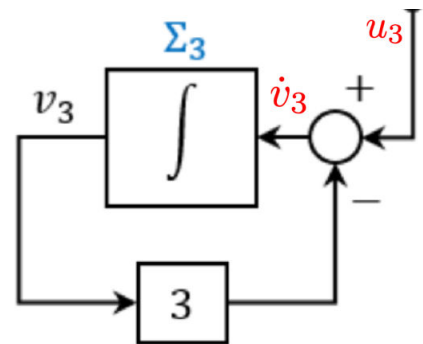


Let's associate the variable $\dot{v}_2$ with the input branch of the "integrator" Σ_2 , as in the figure above (in red in the picture). Moreover, label as input u_2 the branch on the left of the scheme portion, entering in the summation block. Now we are able to write the equations corresponding to this part of the scheme:

$$\begin{cases} \dot{v}_2(t) &= -v_2(t) + u_2(t) \\ y(t) &= v_2(t) \end{cases}$$

Taking into consideration the connection between this part and the previous Σ_1 part of the scheme, it holds: $u_2(t) = y_1(t)$.

Regarding the **last portion of the scheme**,



with a similar approach we obtain the equation

$$\dot{v}_3(t) = -3 v_3(t) + u_3(t)$$

Note: this portion of the scheme has no output branch, so there is no output signal. The input $u_3(t)$ is the output signal of the Σ_2 part of the scheme: $u_3(t) = v_2(t)$

Summing Up

By analysing the scheme, we obtained the following system of differential and algebraic equations:

$$\begin{cases} \dot{v}_1(t) = -2v_1(t) + u(t) \\ \dot{v}_2(t) = -v_2(t) + u_2(t) \\ \dot{v}_3(t) = -3v_3(t) + u_3(t) \\ y_1(t) = v_1(t) \\ u_2(t) = y_1(t) \\ u_3(t) = v_2(t) \\ y(t) = v_2(t) \end{cases}$$

By labeling as **the state variables** x_1, x_2, x_3 the outputs v_1, v_2, v_3 of the integrators (as usual, see the examples in Part-2 of the course material, pp. 12-19), and eliminating the auxiliary variables [i.e. u_2, u_3, y_1 , and y_2], we obtain the desired **state-space realisation**:

$$\begin{cases} \dot{x}_1(t) = -2x_1(t) + u(t) \\ \dot{x}_2(t) = x_1(t) - x_2(t) \\ \dot{x}_3(t) = x_2(t) - 3x_3(t) \\ y(t) = x_2(t) \end{cases}$$

Finally, the matrices A, B, C and D are

$$A = \begin{bmatrix} -2 & 0 & 0 \\ +1 & -1 & 0 \\ 0 & +1 & -3 \end{bmatrix} \quad B = \begin{bmatrix} +1 \\ 0 \\ 0 \end{bmatrix} \quad C = [0 \quad +1 \quad 0] \quad D = [0]$$

- **P3:** Assign the matrices A, B, C and D of the obtained state representation to the **symbolic variables** **symA**, **symB**, **symC** and **symD**.

```
% --- 3rd bullet point
% the matrices A, B, C, and D, according to the state equations

% the matrix symA
symA = sym([-2, 0, 0;
            +1, -1, 0;
            0, +1, -3]);

% the matrix symB
symB = sym([1;0;0]);

% the matrix symC
symC = sym([0, 1, 0]);

% the matrix symD
symD = sym(0);
```

- **P4:** Properly exploiting the Laplace transform, **determine analitically** the **output response** $y(t)$, given the assigned initial conditions and the input function $u(t)$. Store the resulting response in the **symbolic variable** **symYt**.

```
% --- the 4th bullet point
% evaluation of the output movement
```

```
syms s t
```

```
x0 = sym([2.25; 0; -4.81]); % the initial state
```

```
u = 5*exp(-4*t); % the input as symbolic expression
Us = laplace(u,s) % the corresponding Laplace transform
```

$$U_s = \frac{5}{s+4}$$

```
sIA = s*eye(size(symA))-symA; % the symbolic matrix (sI-A)
sIA1 = inv(sIA); % the inverse matrix of (sI-A)

% let's compute the Laplace transform of the output movement
%
% Note: the movement is the summation of the "zero-input" movement
% [or "free movement"] (depending ONLY from the state initial
% condition) and the so-called "zero-state" movement [or
% "forced movement"] (depending ONLY from the input)
%
Ys = simplify(symC*sIA1*x0+symC*sIA1*symB*Us)
```

$$Y_s = \frac{9}{4(s+1)(s+2)} + \frac{5}{(s+1)(s+2)(s+4)}$$

```
% from the Laplace transform to the time-domain variable y(t)
symYt = simplify(ilaplace(Ys,s,t))
```

$$\text{symYt} = \frac{47 e^{-t}}{12} - \frac{19 e^{-2t}}{4} + \frac{5 e^{-4t}}{6}$$

- **P5:** Assign the matrices A , B , C and D of the obtained state representation to the **numeric variables** **matA**, **matB**, **matC** and **matD**.
- **P6:** Using the matrices **matA**, **matB**, **matC** and **matD** define an LTI system description in state-space form, called **stateLTIsys**, exploiting the command **ss()**. **Convert** this representation in a transfer function representation, exploiting the command **tf()**. Assign the obtained transfer function to the variable **tfLTIsys**.

```
% --- the 5th bullet point
% from symbolic to numerical matrices
matA = double(symA);
matB = double(symB);
matC = double(symC);
matD = double(symD);
```

```
% --- the 6th bullet point
% calculate a state-space representation and a I/O reepresentation
% of the LTI system in MATLAB
```

```
% the state-space description
stateLTIsys = ss(matA, matB, matC, matD)
```

```
stateLTIsys =
```

```
A =
      x1  x2  x3
x1  -2   0   0
x2   1  -1   0
x3   0   1  -3
```

```
B =
      u1
x1   1
x2   0
x3   0
```

```
C =
      x1  x2  x3
y1   0   1   0
```

```
D =
      u1
y1   0
```

Continuous-time state-space model.
Model Properties

```
% the transfer function of the same LTI system
tfLTIsys = tf(stateLTIsys)
```

```
tfLTIsys =
```

```
      1
-----
s^2 + 3 s + 2
```

Continuous-time transfer function.
Model Properties

- **P7:** Evaluate the eigenvalues of **matA** and store the result in the numeric array **eigA**.
- **P8:** Evaluate zeros, poles and static gain of **tfLTIsys** and save the results in the **numeric variables zerosTF, polesTF** and **gainTF**.

```
% the 7th bullet point
```

```
% as requested, let's evaluate the eigenvalues
% of the NUMERICAL matrix matA
eigA = eig(matA)
```

```
eigA = 3×1
      -3
      -1
      -2
```



```
% the 8th point  
% calculation of zeros, poles and static gain  
% of the previously computed transfer function  
zerosTF = zero(tfLTIsys)
```

```
zerosTF =  
  
    0×1 empty double column vector
```

```
polesTF = pole(tfLTIsys)
```

```
polesTF = 2×1  
    -2  
    -1
```

```
gainTF = dcgain(tfLTIsys)
```

```
gainTF =  
    0.5000
```